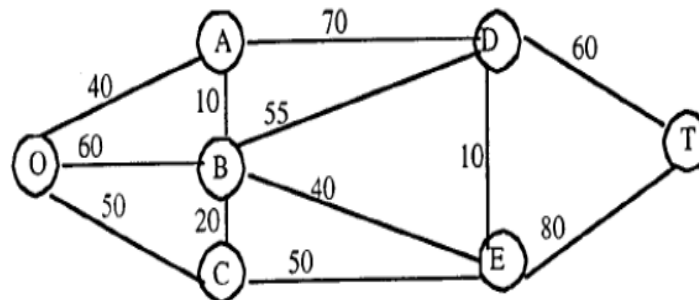


題號 9.3-2 (a)(b)

10.3-2.

(a)



(b)

n	Solved Nodes Directly Connected to Unsolved Nodes	Closest Connected Unsolved Node	Total Distance Involved	n th Nearest Node	Minimum Distance	Last Connection
1	O	A	40	A	40	OA
2, 3	O A	C B	50 40+10 = 50	C B	50 50	OC AB
4	A B C	D E E	40+70 = 110 50+40 = 90 50+50 = 100	E	90	BE
5	A B E	D D D	40+70 = 110 50+55 = 115 90+10 = 100	D	100	ED
6	D E	T T	100+60 = 160 90+80 = 170	T	160	DT

The shortest path from the origin to the destination is $O \rightarrow A \rightarrow B \rightarrow E \rightarrow D \rightarrow T$, with a total distance of 160 miles.

題號 9.3-6 (b)

10.3-6.

(a) The flying times play the role of "distances."

(b) Shortest path: $SE \rightarrow C \rightarrow E \rightarrow LN$, with total flight time 11.3

n	Solved Nodes Directly Connected to Unsolved Nodes	Closest Connected Unsolved Node	Total Distance Involved	n th Nearest Node	Minimum Distance	Last Connection
1	SE	C	4.2	C	4.2	SE-C
2	SE C	A F	4.6 $4.2+3.4 = 7.6$	A	4.6	SE-A
3	SE C A	B F E	4.7 $4.2+3.4 = 7.6$ $4.6+3.4 = 8$	B	4.7	SE-B
4	A B C	E E F	$4.6+3.4 = 8$ $4.7+3.2 = 7.9$ $4.2+3.4 = 7.6$	F	7.6	C-F
5	A B C F	E E E LN	$4.6+3.4 = 8$ $4.7+3.2 = 7.9$ $4.2+3.5 = 7.7$ $7.6+3.8 = 11.4$	E	7.7	C-E
6	A B F E	D D LN LN	$4.6+3.5 = 8.1$ $4.7+3.6 = 8.3$ $7.6+3.8 = 11.4$ $7.7+3.6 = 11.3$	D	8.1	A-D
7	D E F	LN LN LN	$8.1+3.4 = 11.5$ $7.7+3.6 = 11.3$ $7.6+3.8 = 11.4$	LN	11.3	E-LN

7. 找出圖 8.22 之網路的最小擴充樹。

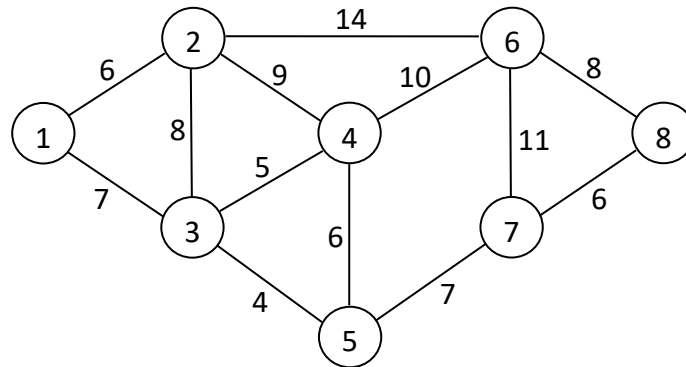


圖 8.22

Sol. 應用最小擴充樹的演算法，可得以下的求解程序（參見圖 f）：

1. 選擇長度最短的弧 (3,5)。
2. 因節點 4 與目前的連接圖最近，所以加入弧 (3,4)。
3. 繼續求解程序，會依序加入 (1,3)、(1,2)、(5,7)、(7,8)、(6,8)。

所得到最小擴充樹的總長度為 43。

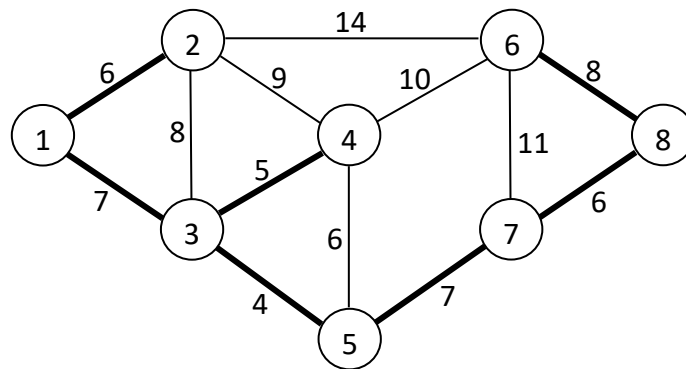


圖 f

14. 考慮圖 8.26 的網路。決定由節點 1 至節點 9 的最大流量以及各弧的最佳流量。

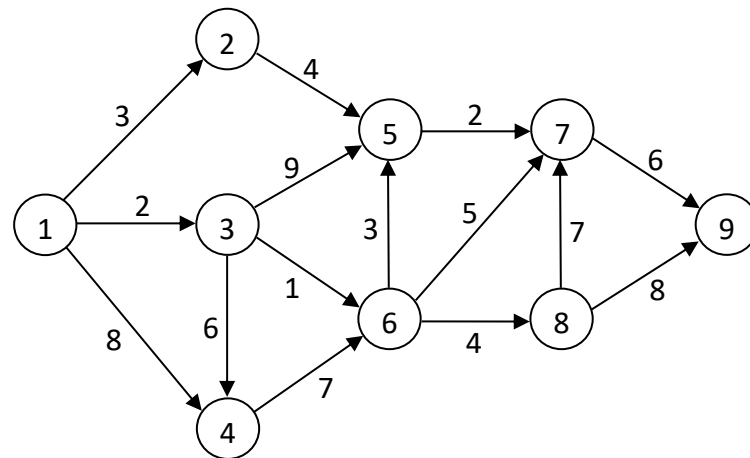


圖 8.26

Sol. 應用最大流量問題演算法可得圖 n 的求解過程。表 G 列出以上所找出的所有路徑。欲決定各弧的最佳流量，我們可比較圖 n 與圖 8.26，兩圖之各弧的流量差異即為各弧的最佳流量，結果如圖 o 所示。

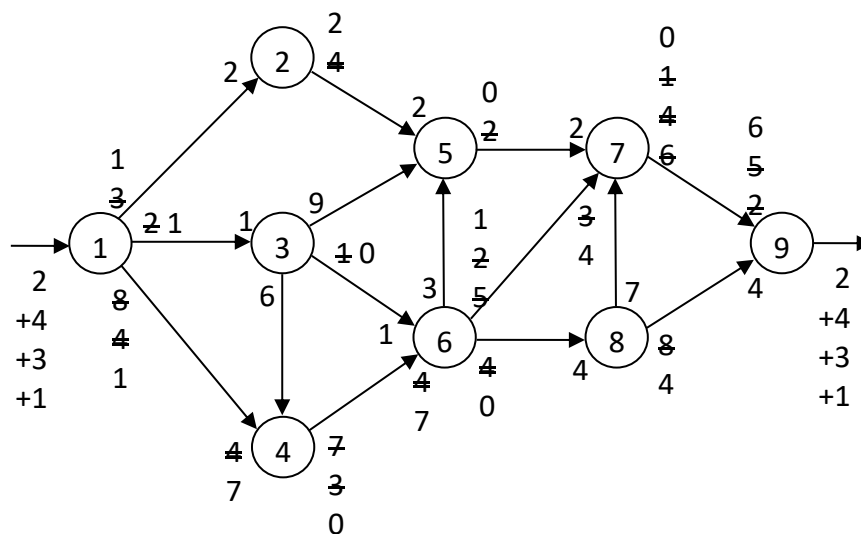


圖 n

表 G

路徑	流量
1 → 2 → 5 → 7 → 9	2
1 → 4 → 6 → 8 → 9	4
1 → 4 → 6 → 7 → 9	3
1 → 3 → 6 → 7 → 9	1
合計	10

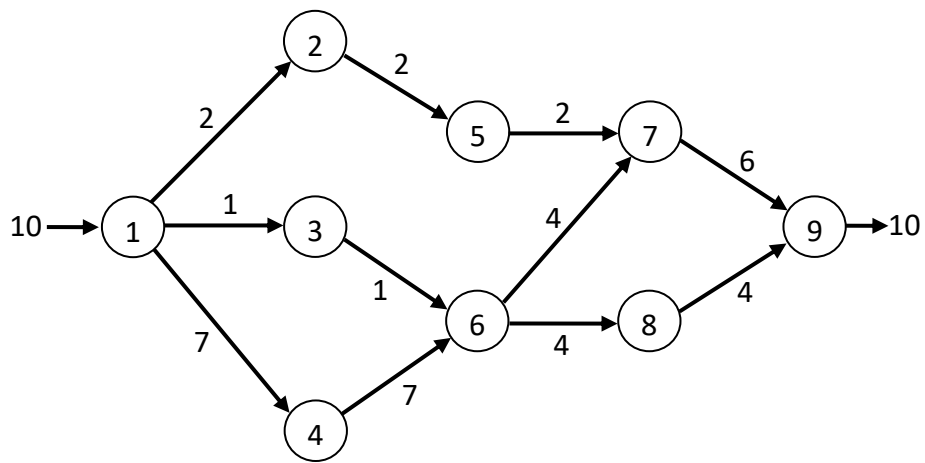
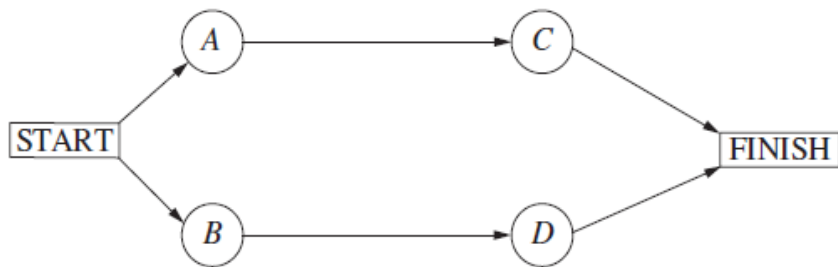


圖 o

題號 9.8-1

注意：課本圖有錯，正確如下：



10.8-1.

Activity to Crash	Crash Cost	Length of Path	
		$A - C$	$B - D$
		14	16
B	\$5,000	14	15
B	\$5,000	14	15
D	\$6,000	14	14
C	\$4,000	13	14
D	\$6,000	13	13
C	\$4,000	12	13
D	\$6,000	12	12

題號 9.8-2 (d)

(d) Let x_j be the reduction in the duration of activity j due to crashing for $j = A, B, C, D$. Also let y_j denote the start time of activity j for $j = C, D$ and y_{FINISH} the project duration.

$$\text{minimize} \quad C = 5000x_A + 5000x_B + 4000x_C + 6000x_D$$

$$\text{subject to} \quad x_A \leq 3, x_B \leq 2, x_C \leq 2, x_D \leq 3$$

$$y_C \geq 0 + 8 - x_A$$

$$y_D \geq 0 + 9 - x_B$$

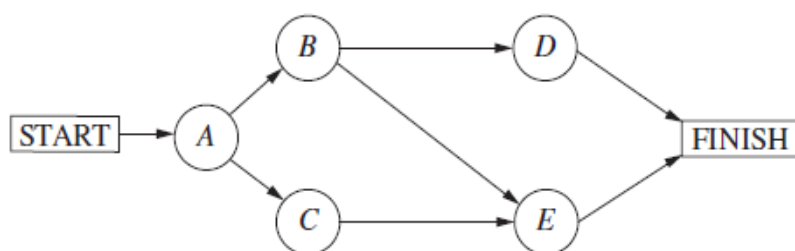
$$y_{\text{FINISH}} \geq y_C + 6 - x_C$$

$$y_{\text{FINISH}} \geq y_D + 7 - x_D$$

$$y_{\text{FINISH}} \leq 12$$

$$\text{and} \quad x_A, x_B, x_C, x_D, y_C, y_D, y_{\text{FINISH}} \geq 0$$

題號 9.8-3 (a) 注意：課本圖有錯，正確如下：



10.8-3.

(a) \$7,834 is saved by the new plan given below.

Activity to Crash	Crash Cost	Length of Path		
		$A - B - D$	$A - B - E$	$A - C - E$
		10	11	12
C	\$1,333	10	11	11
E	\$2,500	10	10	10
$D \& E$	\$4,000	9	9	9
$B \& C$	\$4,333	8	8	8

Activity	Duration	Cost
A	3 weeks	\$54,000
B	3 weeks	\$65,000
C	3 weeks	\$58,666
D	2 weeks	\$41,500
E	2 weeks	\$80,000

→ 68666

(1) 原本成本: $54000 + 62000 + 66000 + 40000 + 75000 + 5000 = 297000$

間接成本: $5000 * 12 = 60000$

原本總成本 $\Rightarrow 297000 + 60000 = 357000$

(2) 現在趕工成本: $54000 + 65000 + 68666 + 41500 + 80000 = 309166$

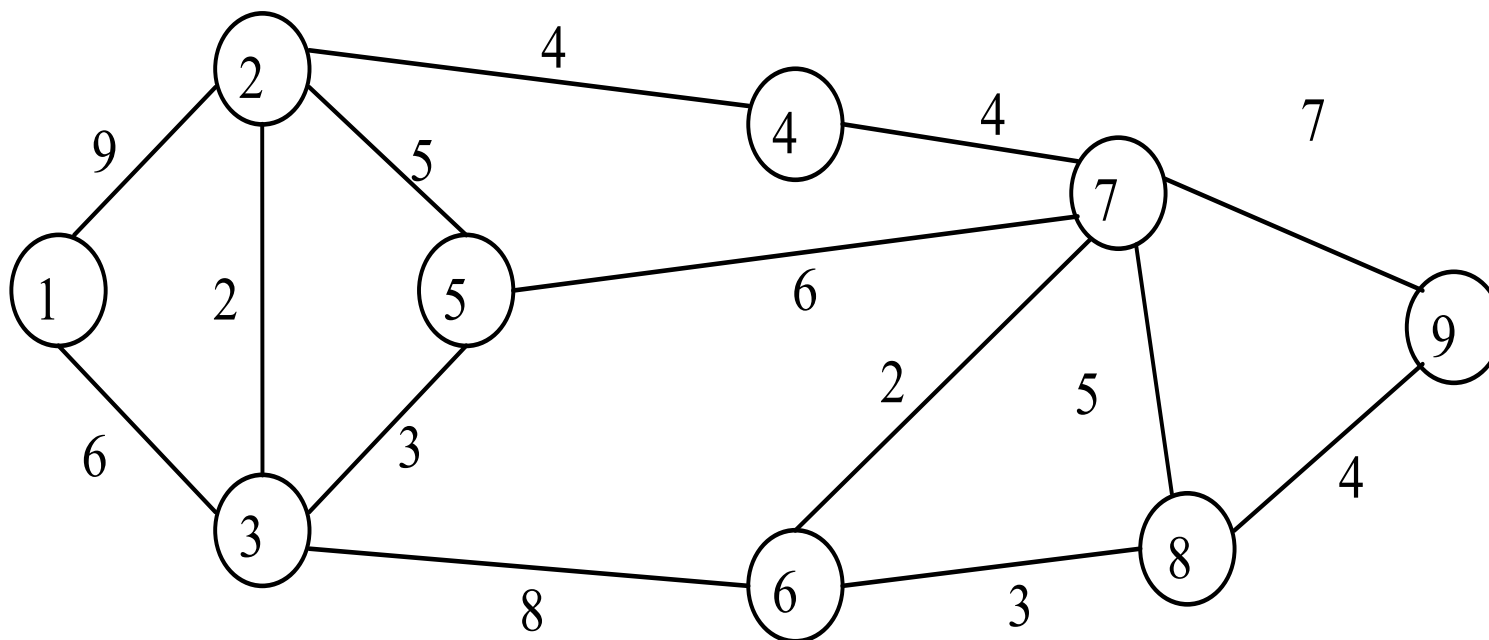
間接成本: $5000 * 8 = 40000$

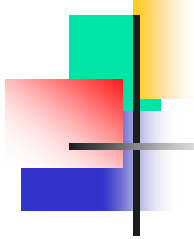
趕工的總成本 $\Rightarrow 309166 + 40000 = 349166$

因此 $\Rightarrow$ 原本總成本 - 趕工的總成本 $= 357000 - 349166 = 7834$

例題8-1

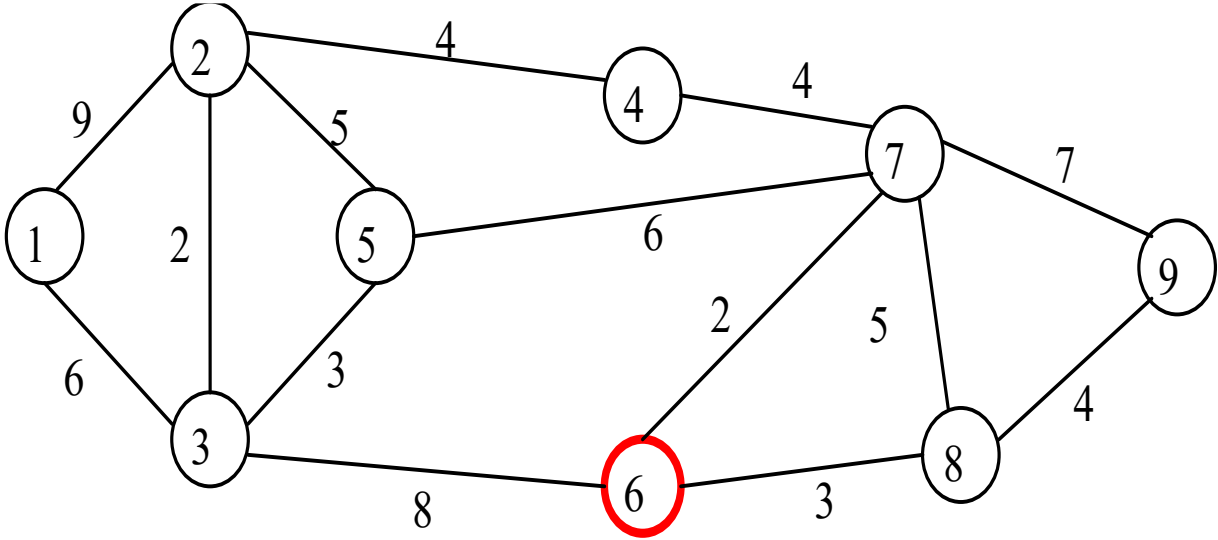
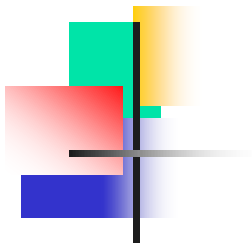
設有網路圖如下，代表各城市間的距離，為架設電腦網路，請求出最短伸展樹



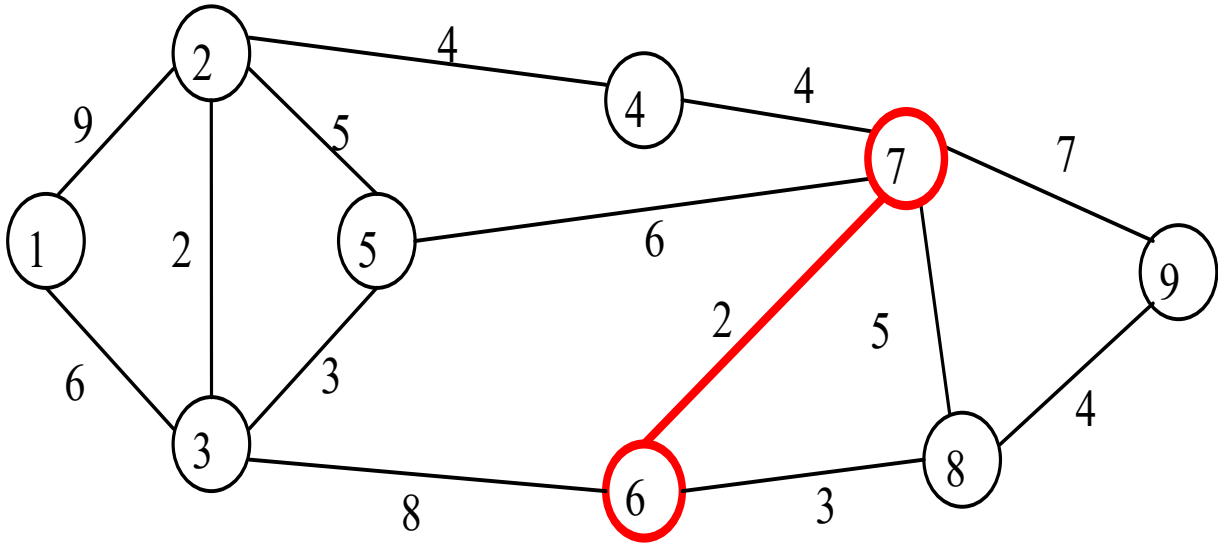


解：

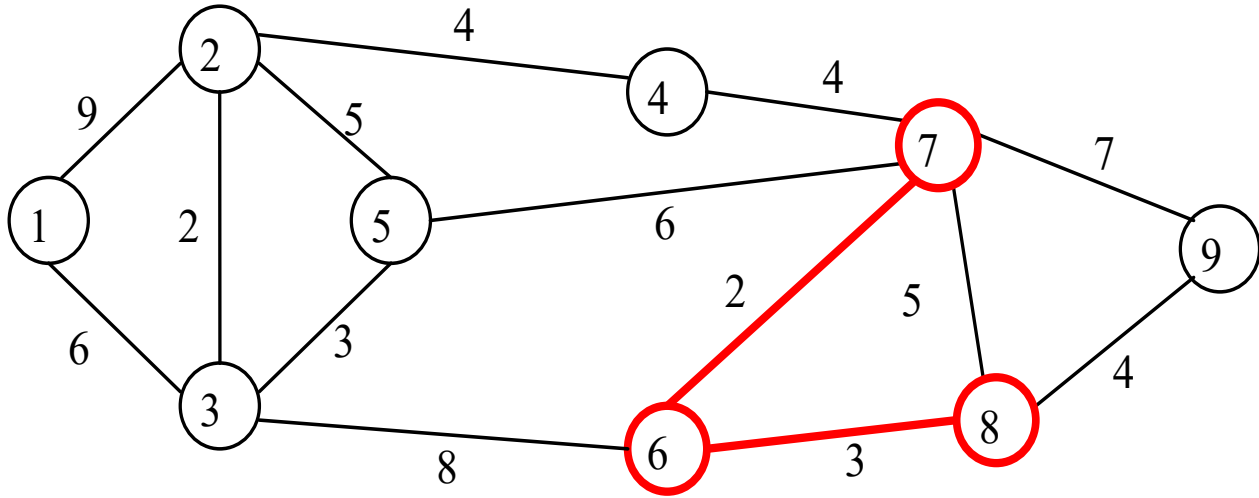
先隨機圈選節點 ⑥。與節點 ⑥ 相連的有節點 ③ ⑦ ⑧，其間弧線以節點 ⑥ ⑦ 間的最短，故再圈選 ⑥ ⑦ 間之弧線與節點 ⑦。再檢視與節點 ⑥ ⑦ 相連的有節點 ③ ④ ⑤ ⑧ ⑨，其間弧線以節點 ⑥ ⑧ 間弧線最短，因而圈選 ⑥ ⑧ 間弧線與節點 ⑧，如此反覆，即可得最佳解，請參見圖 8-4 (a) ~ (I)。



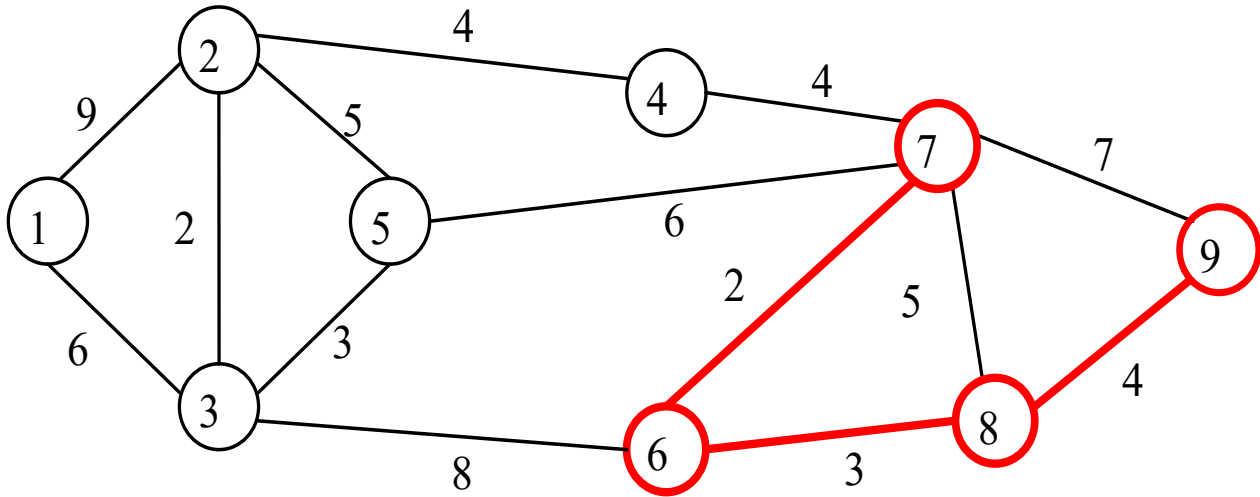
(a)



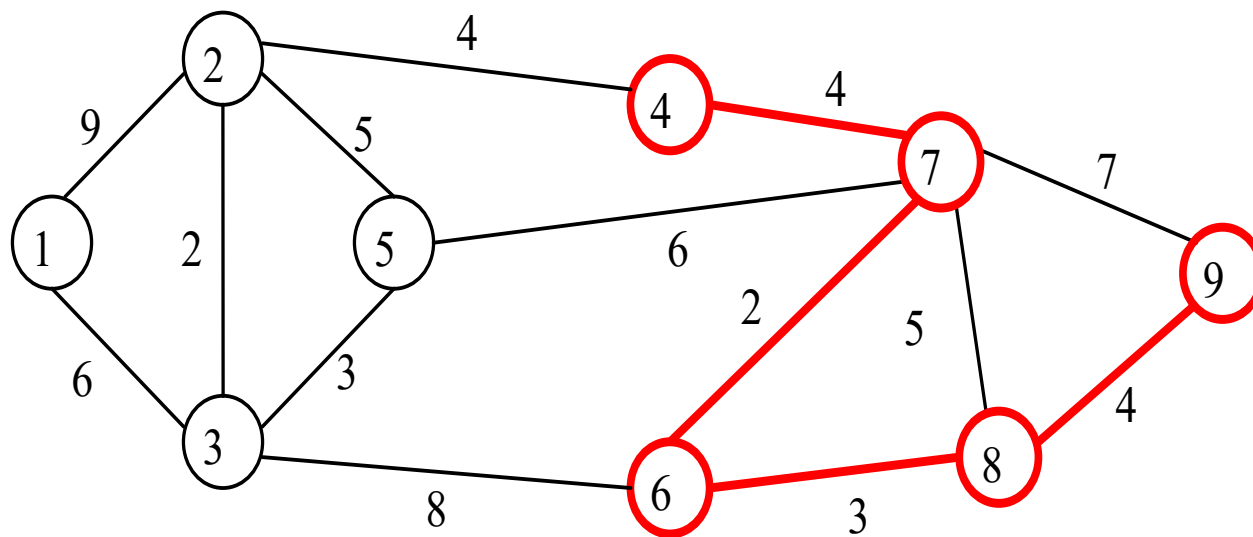
(b)



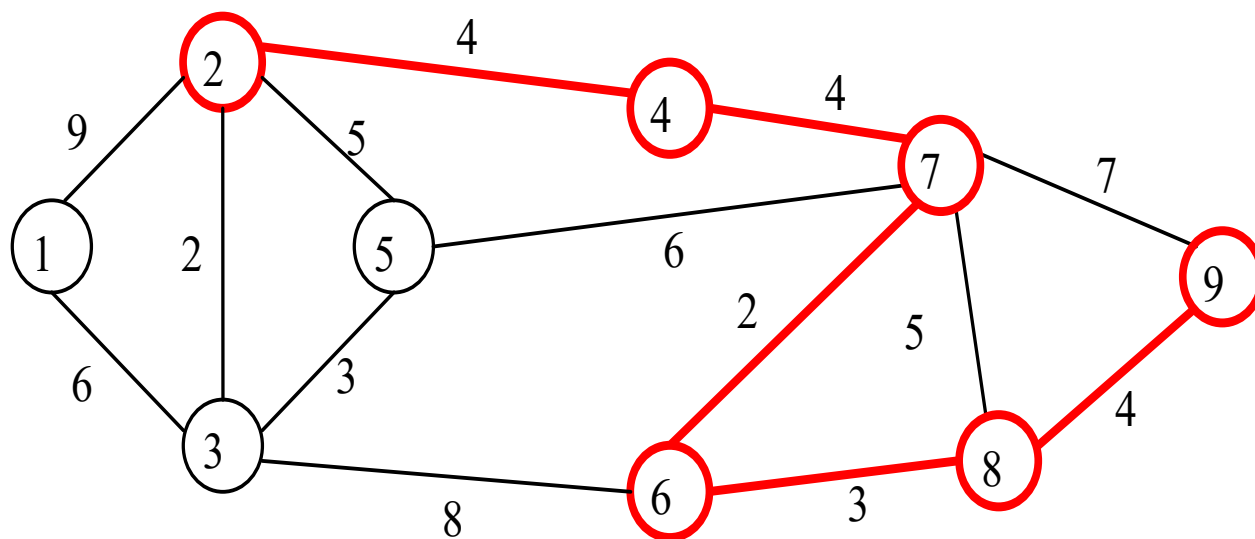
(c)



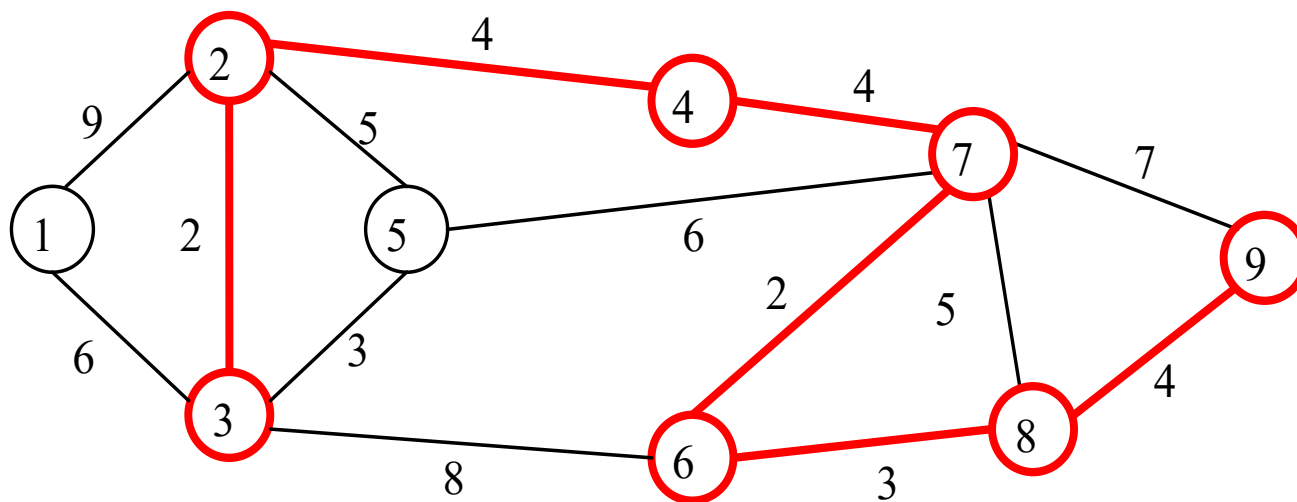
(d)



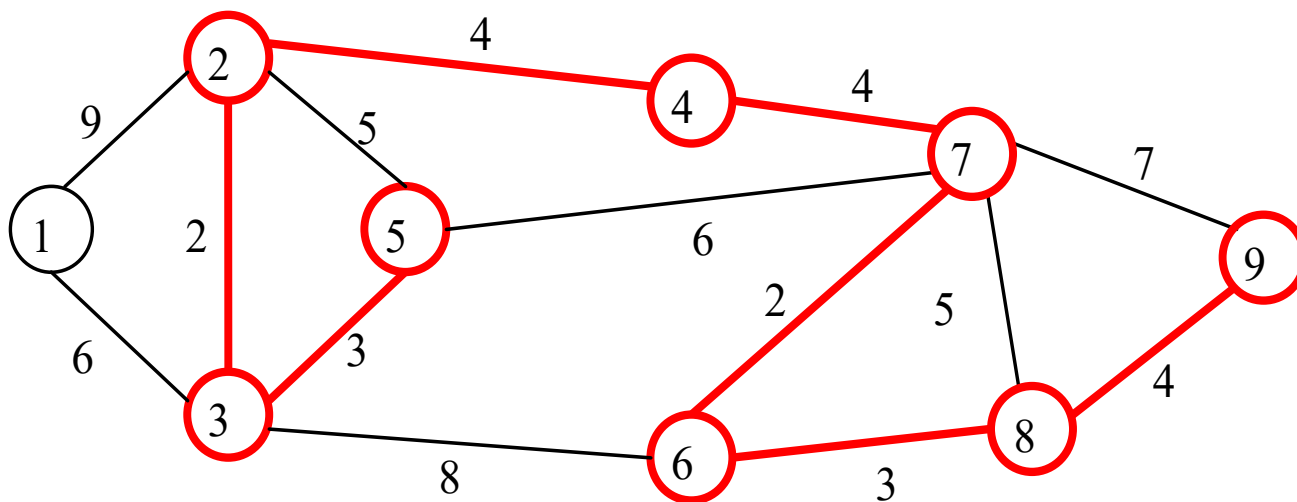
(e)



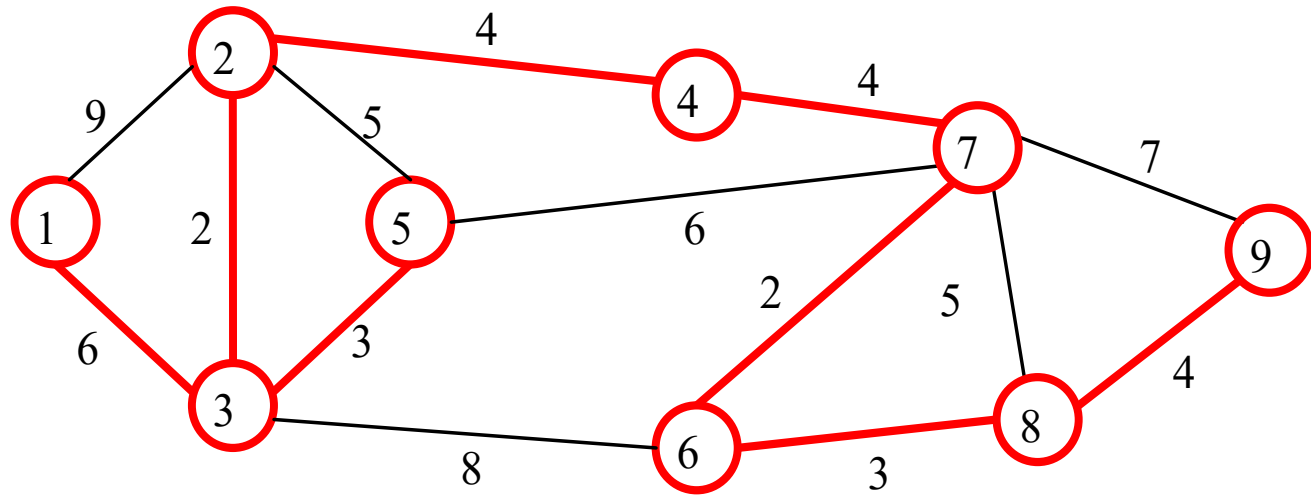
(f)



(g)



(h)

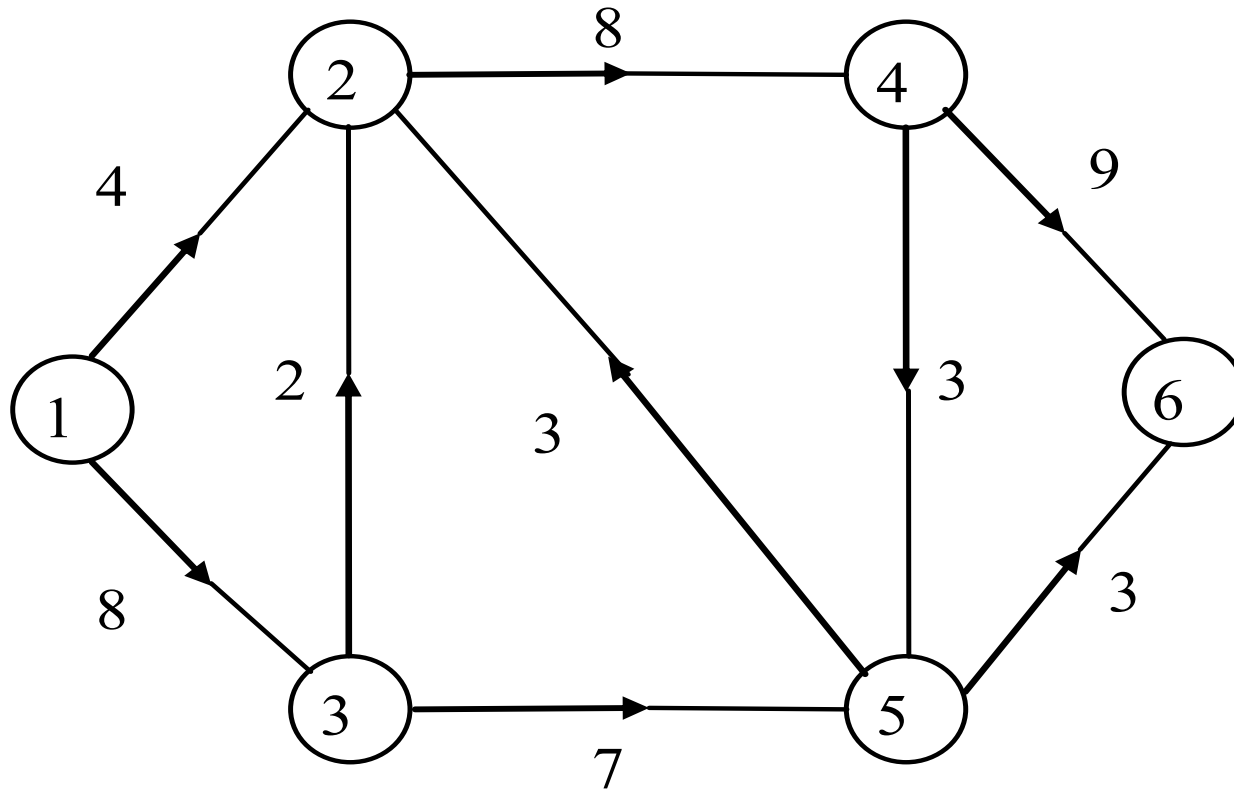


(I)

伸展樹之總長 = $6 + 2 + 3 + 4 + 4 + 2 + 3 + 4 = 28$

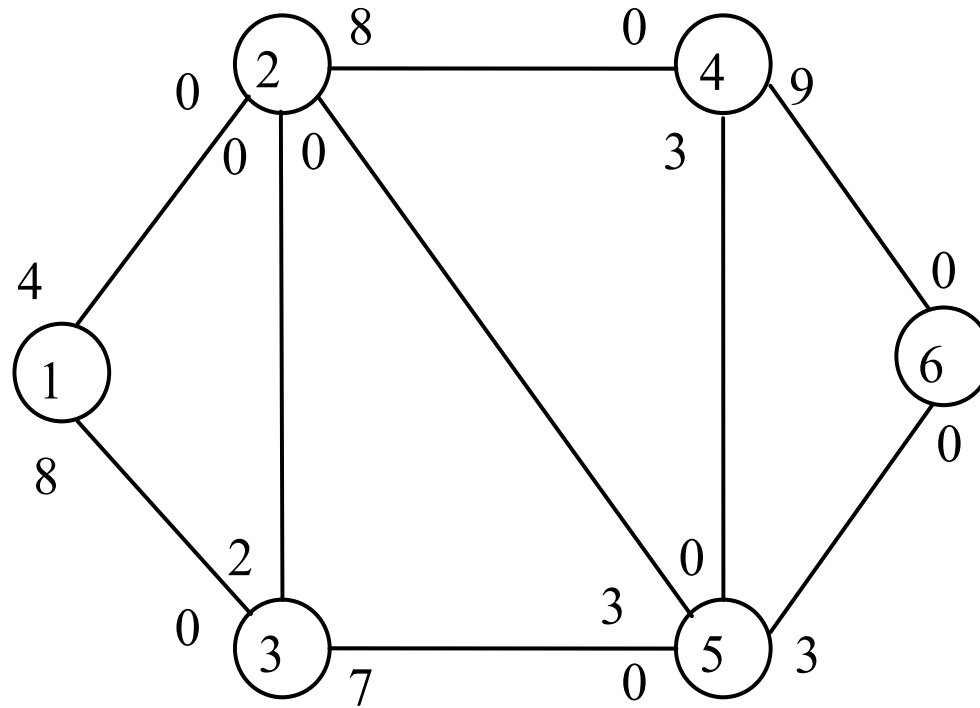
例題8-6

求網路自節點 ① 至節點 ⑥ 的最大流量



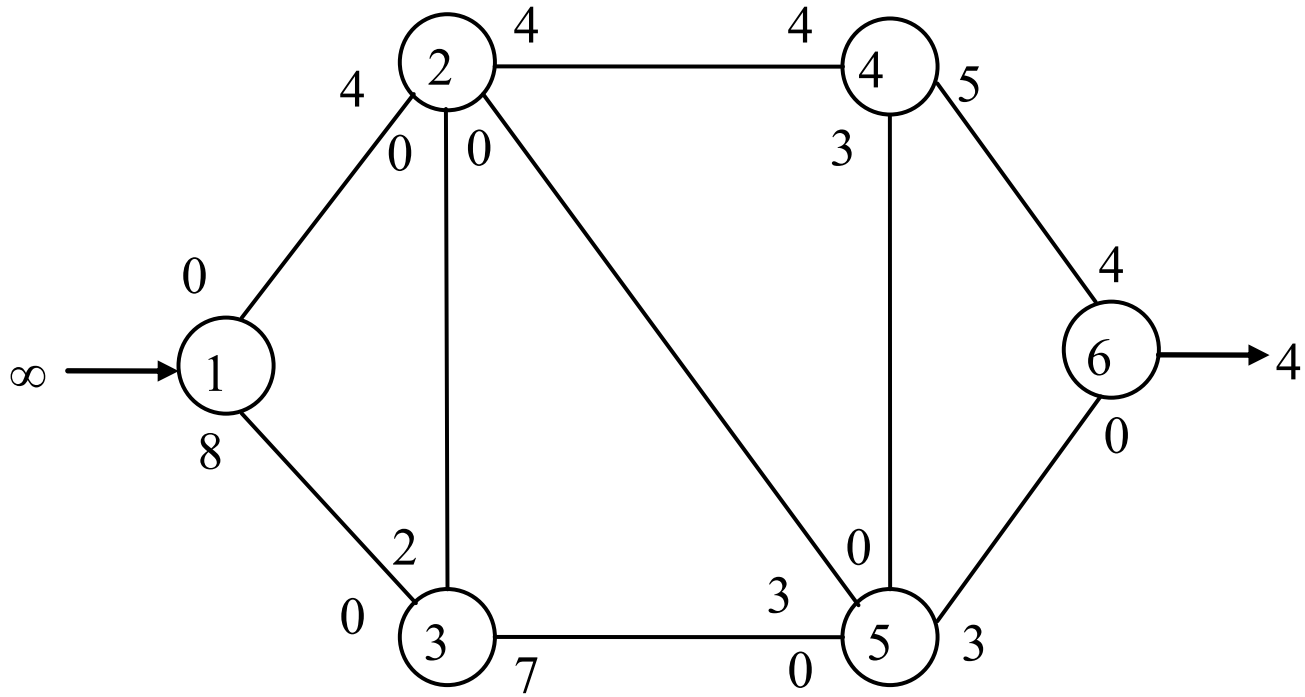
解：

依題目網路圖得下之網路圖。解題過程詳見圖 8-14(a) ~ (e)



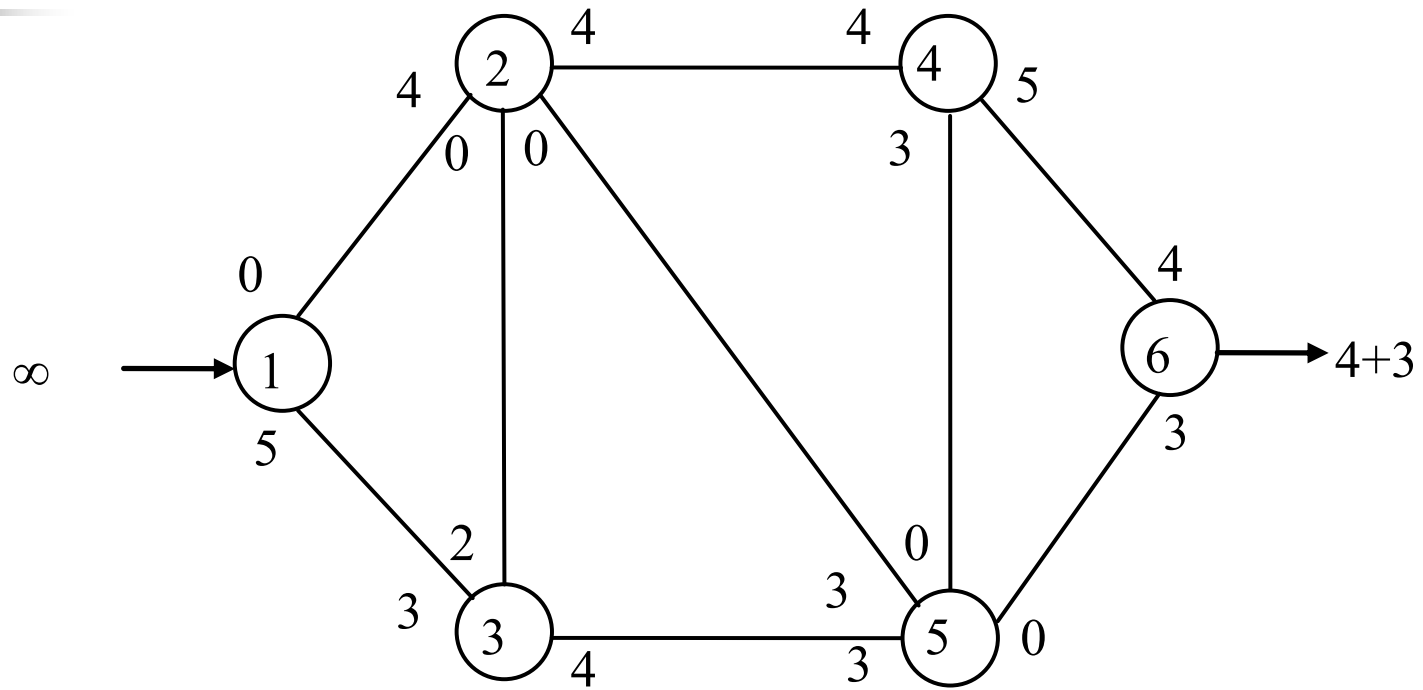
路徑 ① — ② — ④ — ⑥ , $f^* = 4$

(a)



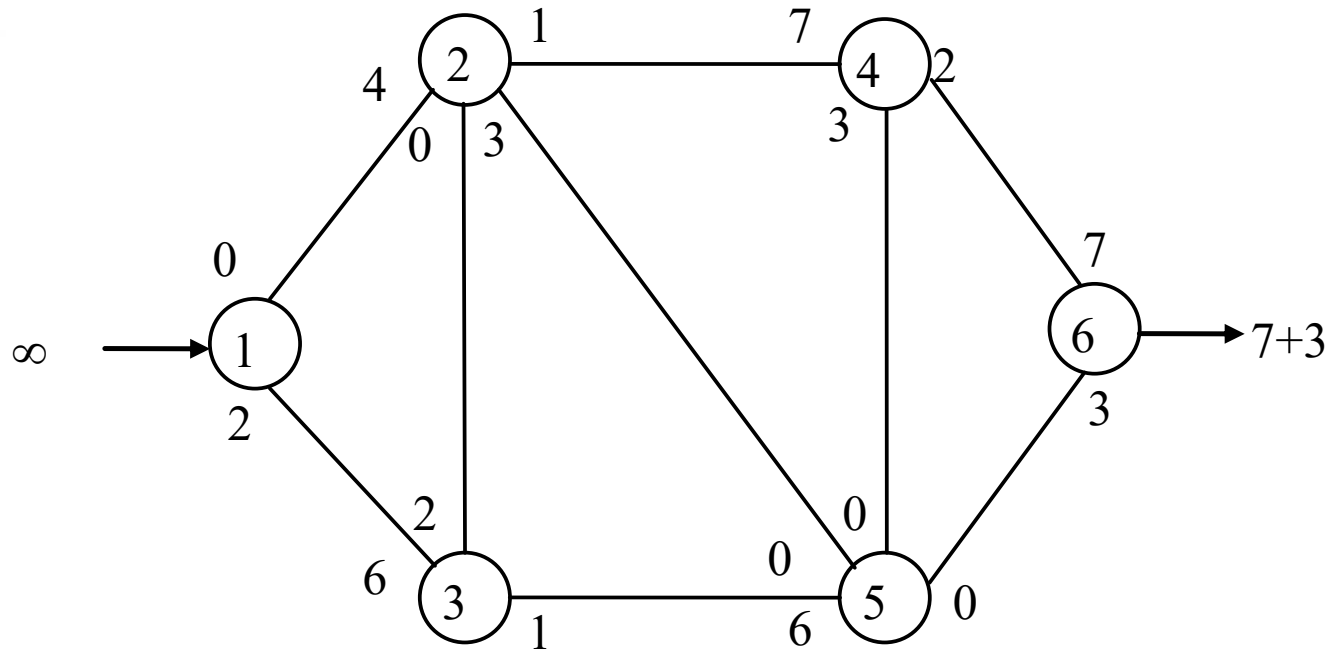
路徑 ① — ③ — ⑤ — ⑥ , $f^*=3$

(b)



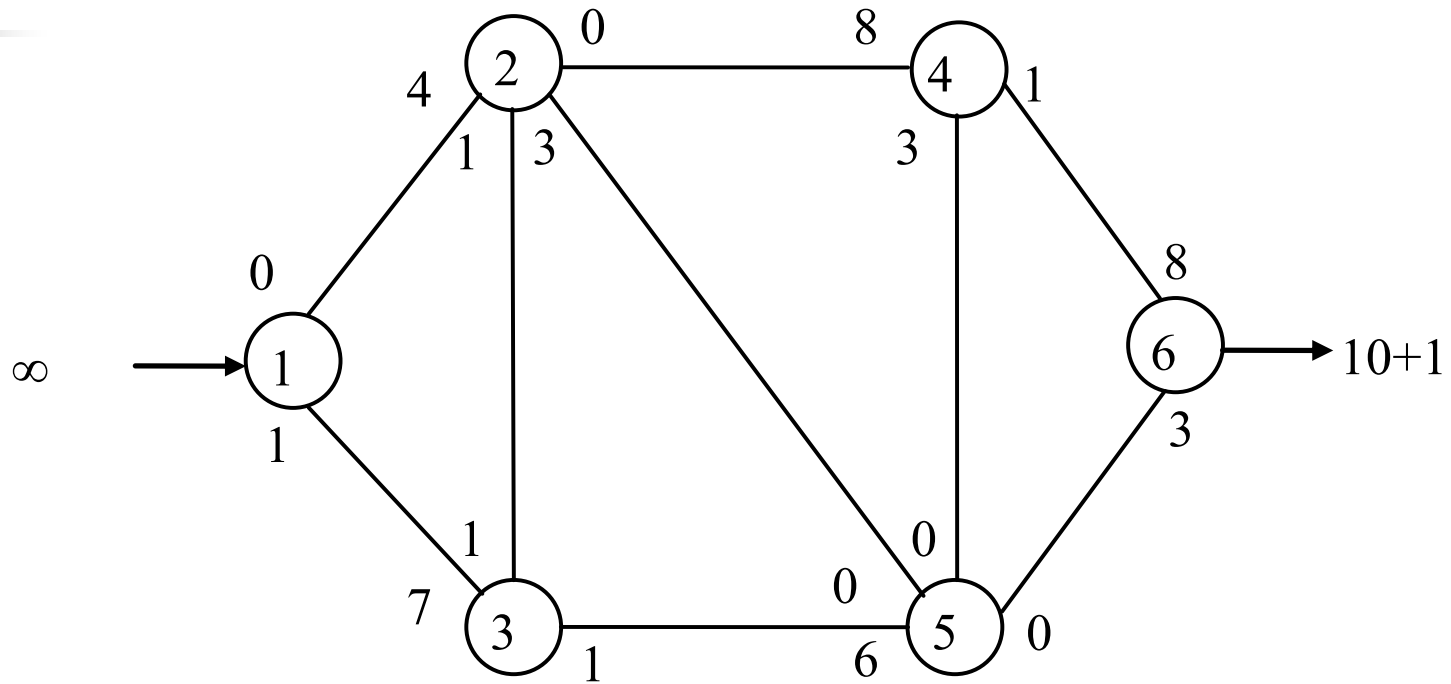
路徑 ① — ③ — ⑤ — ② — ④ — ⑥ , $f^*=3$

(c)



路徑 ① — ③ — ② — ④ — ⑥ , $f^*=1$

(d)



已無自節點 ① 到節點 ⑥ 的正流通量路徑，
故最大流量 11。

(e)